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Studierichting: Informatica

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$$\begin{aligned}\int_{-\infty}^{+\infty} \frac{x}{x^2+4} dx &= \frac{1}{2} \int_{-\infty}^{+\infty} \frac{1}{x^2+4} d(x^2) = \frac{1}{2} \lim_{a \rightarrow -\infty} \int_a^0 \frac{1}{x^2+4} d(x^2) + \frac{1}{2} \lim_{b \rightarrow +\infty} \int_0^b \frac{1}{x^2+4} d(x^2) \\ &= \frac{1}{2} \left(\lim_{a \rightarrow -\infty} [\ln|x^2+4|]_a^0 + \lim_{b \rightarrow +\infty} [\ln|x^2+4|]_0^b \right) \\ &= -\infty + \infty = \text{niet gedefinieerd}\end{aligned}$$

References